XPoint is almost 1000 times faster at reading and writing information than NAND and DRAM.

Intel Optane: This technology is a unique combination of Intel 3D XPoint memory media with Intel-built advanced system memory controller, software IP and interface hardware. Together, these present new computer architecture opportunities and computing possibilities for the market.

Intel Optane represents a new class of memory and storage technology architected specifically for data centre requirements. It offers high capacity, affordability and persistence.

It also provides greater system uptime and faster recovery after power cycles, accelerates virtual machine storage, delivers higher performance to multi-node, distributed cloud applications and offers advanced encryption for persistent data built into the hardware. For memory-intensive workloads, it delivers more server instances at the same service-level agreement performance.

Spin-transfer torque magnetic RAM (STT-MRAM): It stores information in the magnetic states of tiny magnetic elements, or nanomagnets, less than 100 nanometres across. STT-MRAM devices use electric current to read and write data. The technology offers high density, high speed and energy-efficient memory that is non-volatile. With STT, programming current flows through the magnet rather than through write lines adjacent to the bit.

Non-volatile DIMM (NVDIMM): This is essentially a few terabytes of flash or similar storage mounted on a DIMM. Because it is on a fast bus and uses memory IO methods, NVDIMM is four times faster than the fastest SSD. It creates a journal file of write transactions and maintains an image of the data in-memory, without affecting operations.

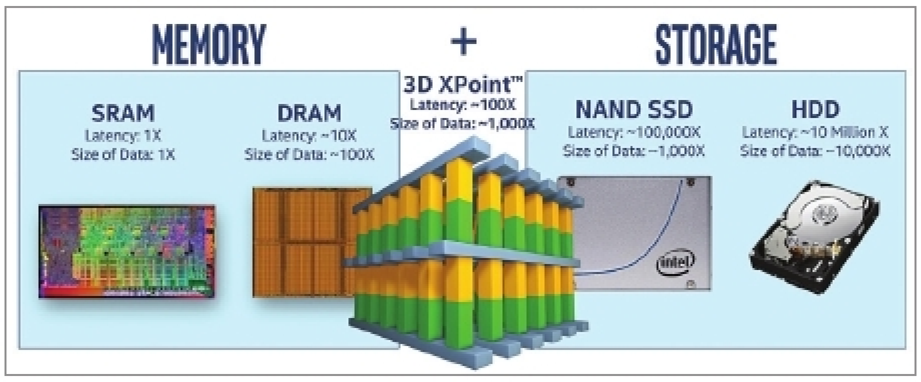

Figure 2: 3D XPoint memory technology (Credit: *https://arstechnica.com*)

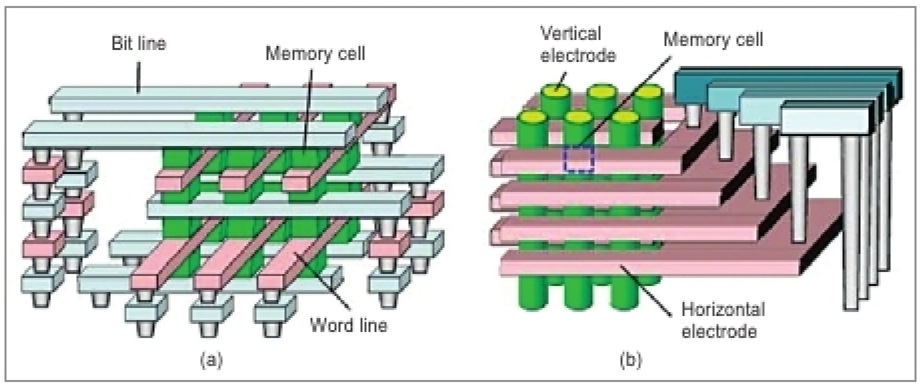

Figure 3: A non-linear ReRAM cell with sub-1μA ultra-low operating current for high density vertical resistive memory (Credit: *www.semanticscholar.org*)

In-memory processing infrastructure

In-memory processing infrastructure is evolving rapidly. And, the most interesting potential changes are for NVDIMM and CPU architectures. For example, Intel's Optane technology has byte addressability. This enables it to write a single byte to Optane memory with a CPU register using memory command. It first creates a block in the app and then goes through the storage stack software. Finally, it transmits a minimum of 4kB with a single CPU operation. This complete process takes one or two machine cycles.

Network bandwidth and local storage are the key bottlenecks in Big Data environments. The solution is data compression, where compression ratio depends on data—typically, it reduces up to five times.

Another method is to increase DRAM capacity and performance with a much tighter coupling of CPU and memory. Several initiatives have been taken based on hybrid memory cube concept to improve the performance of memories. The aim is to achieve lower power consumption by memories and higher memory speed. Hybrid DRAM, flash and CPU module architectures speed up in-memory operations significantly.

Emerging memories include MRAM, resistive RAM (ReRAM), PCM and ferroelectric RAM (FeRAM). MRAM needs to be integrated into complementary metal oxide semiconductor production lines to achieve higher capacities. These can be used as on-chip memories that allows larger memories for multi-core architecture.

For advanced memory storage, ReRAM can be used for applications such as image recognition and AI.

NVM, PCM, STTRAM, ReRAM and 3D XPoint have larger capacity, higher latency, higher dynamic power and lower endurance. 3D, JBOF and JBOD technologies are also revolutionising memories. The main memory system must scale in technology, cost, efficiency, size and management algorithms to maintain performance growth and technology-scaling benefits. Decisions of which technology to use must be based on required performance and total cost of the system. END

Differences between latest non-volatile memories

Non-volatile memory	Working principle
PCM	• Injects current to change material phase • Resistance determined by phase
STT-MRAM	• Injects current to change magnet polarity • Resistance determined by polarity
Memristors/RRAM/ReRAM	• Injects current to change atomic structure • Resistance determined by atom distance

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By: EFY Bureau